Tutorial-7

KNN - K-nearest neighbour

$$d_{T_i} = \sqrt{\sum_{j=1}^{d} (\text{Feature}_j^T - \text{Feature}_j^i)^2}$$

d_{T_i} → distance from target to sample i

$\sum_{j=1}^{d}$ → Summation over all the features for sample i.

$$d_{T_1-1} = \sqrt{(20-25)^2 + (6-8)^2 + (3-5)^2 + (90-126)^2}$$

$$= \sqrt{(-5)^2 + (-2)^2 + (-2)^2 + (36)^2}$$

$$= \sqrt{25 + 4 + 4 + 900} = \sqrt{933} = 30.5450$$

$$d_{T_2} = \sqrt{(20-5)^2 + (6-2)^2 + (3-0)^2 + (90-6)^2}$$

$$= \sqrt{(15)^2 + (4)^2 + 3^2 + (90)^2}$$

$$= \sqrt{8350} = 91.3783$$

$$d_{T_1-3} = \sqrt{(20-30)^2 + (6-10)^2 + (3-7)^2 + (90-150)^2}$$

$$= \sqrt{3732} = 61.0900$$

$$d_{T_1-4} = \sqrt{(20-8)^2 + (6-3)^2 + (3-1)^2 + (90-20)^2}$$

$$= \sqrt{(12)^2 + (3)^2 + (2)^2 + (50)^2}$$

$$= \sqrt{5057} = 71.1125$$

$$d_{T_1-5} = \sqrt{(20-22)^2 + (6-7)^2 + (3-4)^2 + (90-100)^2}$$

$$= \sqrt{4 + 1 + 1 + 100} = \sqrt{106} = 10.2956$$

$$d_{T_1-6} = \sqrt{(20-6)^2 + (6-2)^2 + (3-0)^2 + (90-15)^2}$$

$$= \sqrt{5846} = 76.4591$$

$d_{T_1-7} = \sqrt{(20-28)^2 + (6-9)^2 + (3-6)^2 + (90-140)^2}$

$= \sqrt{2582}$

$= 50.8133$

$d_{T_1-8} = \sqrt{(20-7)^2 + (6-3)^2 + (3-1)^2 + (90-25)^2}$

$= \sqrt{4407} = 66.3852$

<u>For K = 1</u>

$d_{T_1-5} \rightarrow Y$

<u>For K = 3</u>

$\overset{Y}{d_{T_1-5}}, \overset{Y}{d_{T_1-1}}, \overset{Y}{d_{T_1-7}} \rightarrow Y$ (class)

<u>for K = 5</u>

$\overset{Y}{d_{T_1-5}}, \overset{Y}{d_{T_1-1}}, \overset{X}{d_{T_1-7}}, \overset{Y}{d_{T_1-3}}, \overset{N}{d_{T_1-8}}$

$= Y \rightarrow$ max is Y

Confusion Matrix

	Pred Y	Predict N
Actual Y	TP	FN
Actual N	FP	TN

$T_{2-1} = \sqrt{(10-25)^2 + (4-8)^2 + (2-5)^2 + (40-120)^2}$

$= \sqrt{6650} = 81.5475$

$T_{2-2} = \sqrt{(10-5)^2 + (4-2)^2 + (2-0)^2 + (40-0)^2}$

$= \sqrt{1633} = 40.4103$

$T_{2-3} = \sqrt{(10-30)^2 + (4-10)^2 + (2-7)^2 + (40-150)^2}$

$= \sqrt{12561} = 112.0758$

$T_{2-4} = \sqrt{(10-8)^2 + (4-3)^2 + (2-1)^2 + (40-20)^2}$

$= \sqrt{406} = 20.1494.$

$T_{2-5} = \sqrt{(10-22)^2 + (4-7)^2 + (2-0)^2 + (40-100)^2}$

$= \sqrt{3757} = 61.2943.$

$T_{2-6} = \sqrt{(10-6)^2 + (4-2)^2 + (2-0)^2 + (40-15)^2}$

$= \sqrt{649} = 25.4754.$

$T_{2-7} = \sqrt{(10-28)^2 + (4-9)^2 + (2-6)^2 + (40-140)^2}$

$= \sqrt{10365} = 101.8086.$

$T_{2-8} = \sqrt{(10-7)^2 + (4-3)^2 + (2-1)^2 + (40-25)^2}$

$= \sqrt{236} = 15.3622$ //

Sorted Distance Order:-

T_{2-8}, T_{2-4}, T_{2-6}, T_{2-2}, T_{2-5}, T_{2-1}, T_{2-7}, T_{2-3}

For $K = 1$;

$T_{2-8} = N$ //

For $K = 3$:

$T_{2-8} \rightarrow N$, $T_{2-4} \rightarrow N$, $T_{2-6} \rightarrow N \Rightarrow N$ //

For $K = 5$:

$T_{2-8} \rightarrow N$; $T_{2-4} \rightarrow N$, $T_{2-6} \rightarrow N$, $T_{2-2} \rightarrow N$,

$T_{2-5} \rightarrow Y \Rightarrow N$ //

$T_{3-1} = \sqrt{(27-25)^2 + (9-8)^2 + (5-5)^2 + (130-120)^2}$

$= \sqrt{105} = 10.2.$

$T_{3-2} = \sqrt{(27-5)^2 + (9-2)^2 + (5-0)^2 + (130-0)^2}$

$= \sqrt{17458} = 132.1$

$T_{3-3} = \sqrt{(27-30)^2 + (9-10)^2 + (5-7)^2 + (130-150)^2}$

$= \sqrt{414} = 20.3$

$T_{3-4} = \sqrt{(27-8)^2 + (9-3)^2 + (5-1)^2 + (130-20)^2}$

$= \sqrt{12513} = 111.86$

$T_{3-5} = \sqrt{(27-22)^2 + (9-7)^2 + (5-4)^2 + (130-100)^2}$

$= \sqrt{930} = 30.4$

$T_{3-6} = \sqrt{(27-6)^2 + (9-2)^2 + (5-0)^2 + (130-15)^2}$

$= \sqrt{15740} = 117.2$

$T_{3-7} = \sqrt{(27-28)^2 + (9-9)^2 + (5-6)^2 + (130-140)^2}$

$= \sqrt{102} = 10.09$

$T_{3-8} = \sqrt{(27-7)^2 + (9-3)^2 + (5-1)^2 + (130-25)^2}$

$= \sqrt{11477} = 109.13$

Sorted Distance Order:

$T_{3-7}, T_{3-1}, T_{3-3}, T_{3-5}, T_{3-8}, T_{3-4}, T_{3-2}$

For $k=1$: $T_{3-7} \rightarrow Y$

For $K=3$: $T_{3-7} \rightarrow Y$, $T_{3-1} \rightarrow Y$, $T_{3-5} \rightarrow Y \Rightarrow Y$

For $K=5$: $T_{3-7} \rightarrow Y$, $T_{3-1} \rightarrow Y$, $T_{3-5} \rightarrow Y$,

$T_{3-8} \rightarrow N \Rightarrow Y$

For K = 1 Accuracy = 100 %.

For K = 3 Accuracy = 100 %.

For K = 5 Accuracy = 100 %.

Confusion Matrix

K = 3.

	Pred Y	Predict. ~~Actual~~ N
Actual Y	2	0
Actual N	0	1.